INFLUENCE OF CRYSTALLINE ADMIXTURE, ECBP AND PVA FIBER ON THE SELF-HEALING PERFORMANCE OF CONCRETE

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ABSTRACT

The durability of concrete structures can be significantly enhanced through the incorporation of self-healing mechanisms. This study evaluates the influence of crystalline admixture (CA), Electrical Circuits Boards Powder (ECBP), and polyvinyl alcohol fiber (PVA) on the self-healing behavior of concrete containing early-age cracks. Four concrete mixtures were prepared, including a control mix, concrete with 0.1% PVA fiber by volume, concrete with 20% cement replacement by ECBP, and concrete containing 2% crystalline admixture by mass of cement. Early-age cracks were induced after three days of curing using controlled compressive loading. Self-healing performance was assessed through compressive strength recovery, electrical resistivity recovery, and microstructural examination using scanning electron microscopy (SEM). The results indicate that all modified concrete mixes exhibited measurable self-healing behavior; however, the concrete incorporating crystalline admixture demonstrated superior recovery of both mechanical and durability-related properties. SEM analysis confirmed that the enhanced healing efficiency was primarily attributed to the formation of calcium carbonate and calcium silicate hydrate (C–S–H) gel within the crack zones.

KEYWORDS

Concrete ; crystalline admixture (CA) ; Electrical Circuits Boards Powder (ECBP) ; polyvinyl alcohol fiber (PVA) ; early-age ; self healing ; crackscompressive strength ; electrical resistivity.

1 Introduction

Concrete and cement-based materials remain indispensable in civil engineering, and their complete replacement is unlikely in the near future. Cracking in concrete structures is unavoidable due to the material's low tensile strength relative to its compressive capacity . Such cracks increase permeability and facilitate the ingress of aggressive agents, including chlorides and carbon dioxide, which accelerate carbonation, reinforcement corrosion, and deterioration of structural durability. If left untreated, cracking may impair serviceability and ultimately lead to structural failure . Consequently, there is growing interest in developing concrete systems capable of restoring mechanical performance and reducing permeability after early-age cracking .

Self-healing in concrete primarily occurs through the precipitation of calcium carbonate formed by the reaction of calcium ions with dissolved carbon dioxide in the presence of moisture, as well as through the continued hydration of unhydrated cement particles. This

process, known as autogenous healing , is most effective for narrow cracks and can be enhanced by the inclusion of lime, pozzolanic materials, and mineral admixtures . In general, self-healing mechanisms in concrete are classified as autogenous or autonomous healing systems .

1.1 Research significance

This study evaluates the effects of crystalline admixture, ECBP, and PVA fiber on the self-healing of early-age cracks in concrete based on mechanical and durability recovery and crack closure. Four concrete mixes were comparatively assessed under water immersion, demonstrating enhanced self-healing potential for water-tight applications.

2 Experimental program and methodology

2.1 Experimental program

Self-healing performance was evaluated through the recovery of mechanical and durability properties. After 3 days of water curing, specimens were pre-cracked under compression at an optimum age to produce stable and well-defined cracks without crushing or brittle failure . Following cracking, specimens were immersed in water for 42 days prior to mechanical and durability testing, exceeding the healing durations commonly reported in previous studies.

2.2 Materials and mix proportions

In this study ordinary Portland cement (OPC), ECBP, superplasticizer, water, coarse aggregate, fine aggregate, crystalline admixture (CA), and polyvinyl alcohol (PVA) fibers were used. OPC of 43 grade was selected. Coarse aggregate with a nominal size of 10–12 mm and river sand as fine aggregate were employed. Four concrete mixes were prepared: control (M1), 0.1% PVA fiber by volume (M2), 20% cement replacement with ECBP (M3), and 2% crystalline admixture by mass of cement (M4). Mix proportions and material properties are reported in Tables 1 and 2. Concrete cube specimens (100 mm × 100 mm × 100 mm) were cast for mechanical and durability evaluation (Fig. 1).

2.3 Experimental program

Self-healing behavior was evaluated through a defined experimental sequence involving damage induction, water curing, and post-healing assessment of recovered properties . The methodology consisted of four stages: early-age crack formation, evaluation of mechanical and durability recovery, crack surface characterization, and microstructural analysis.

2.3.1 Creation of early age cracks in the specimens

Cube specimens were pre-cracked at 3 days using a compression testing machine to induce early-age cracking. A constant loading rate of 20 kN/min was applied until the first visible crack formed, with crack opening limited to approximately 0.4 mm under visual control . As crack width could not be precisely controlled during loading, variations in crack width were observed among mixes . Surface crack widths were measured using a crackscope instrument. Specimen geometry and material elemental composition are presented in Fig. 1 and Table 2, respectively.

Table1 Concrete mix proportions (kg/m^3)

Mixes	Cement	Fine aggregate	Coarse aggregate	Water Content (lit/m3)	Super-plasticizer (lit/m3)	ECBP	PVA fiber	Crystalline admixture	Slump (mm)
Control (M1)	413	907.69	956.56	165.36	3.307	0	0	0	36
PVA 2 (M2)	413.40	906.39	955.19	165.36	3.307	0	1.3	0	27
ECBP 20 (M3)	363.79	874.75	921.85	165.36	3.307	90.95	0	0	44
CA 2 (M4)	413.40	903.85	952.52	165.36	3.307	0	0	8.27	31

Table 2 Elemental contents of OPC , ECBP and CA

Chemicals	Elemental contents (%)		
	OPC	ECBP	CA
Silica(S)	14.26	25.2	12.37
Calcium	68.78	40.4	68.72
Aluminium (Al)	3.98	27.9	9.32
Iron (Fe)	3.97	0.463	2.95
Magnesium (Mg)	1.98	0.896	7.02
Sulphur trioxide (K)	3.24	0.364	1.45
Soda (Na2O)+Potash(K2O)	3.29	0.963	3.08

Fig.1Concrete specimen

Fig. 2. Pre-cracking process of concrete specimen

2.3.2 Assessment of recovery of mechanical and durability properties of cracked concrete

Self-healing performance was quantified using the compressive strength recovery ratio $R(s)$and electrical resistivity recovery ratio $R(r)$, calculated according to Eqs. (1) and (2).

$$R(s) = \frac{\text{Compressive strength after healing}}{\text{Original compressive strength}} \times 100 \qquad (1)$$

$$R(r) = \frac{\text{Electrical resistivity after healing}}{\text{Original electrical resistivity}} \times 100 \qquad (2)$$

3. Results and Discussion

3.1. Recovery of Mechanical Properties

Compressive strength recovery was evaluated for concrete specimens pre-cracked at 3 days and subjected to 42 days of water curing . Four mixes were examined: control concrete (M1), concrete with 0.1% PVA fiber by volume (M2), concrete with 20% ECBP replacement (M3), and concrete with 2% crystalline admixture by mass of cement (M4).

The compressive strength recovery ratios relative to the corresponding uncracked specimens were 78.51% (M1), 84.37% (M2), 85.95% (M3), and 93.98% (M4), as shown in Fig. 4. The highest recovery was observed in mix M4. Strength recovery is associated with microcrack healing resulting from continued hydration, C–S–H gel formation, and pore-blocking products.

3.2. Recovery of Durability Properties

Bulk electrical resistivity was determined in accordance with ASTM C1760-12 using a resistivity meter (Fig. 5). After the healing period, wet cube specimens were placed between two parallel metal plates with moist sponge contact. An alternating current of the specified frequency was

applied, and the voltage across the specimen was recorded.

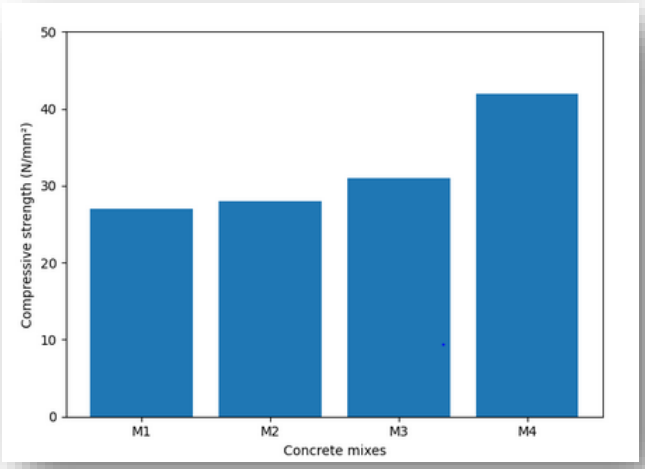


Fig. 3. Compressive strength of cracked concrete (42 days) (N/mm2).

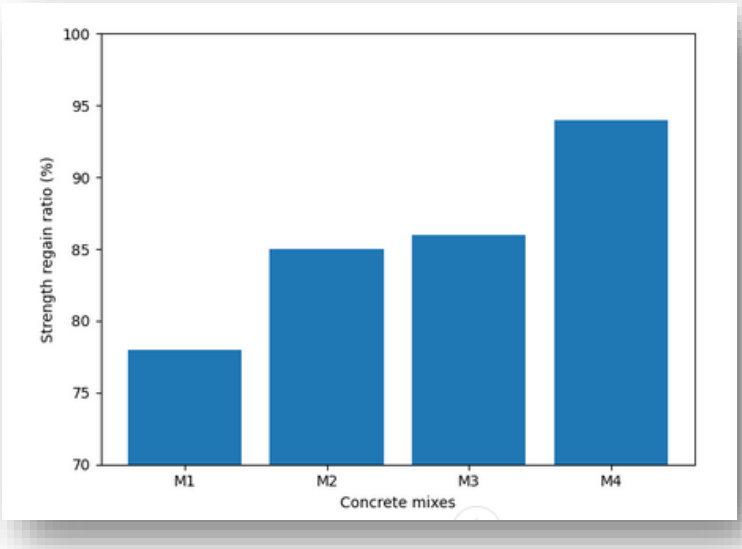


Fig. 4. Strength recovery ratio R(s) (%).

Fig. 5. Electrical resistivity meter.

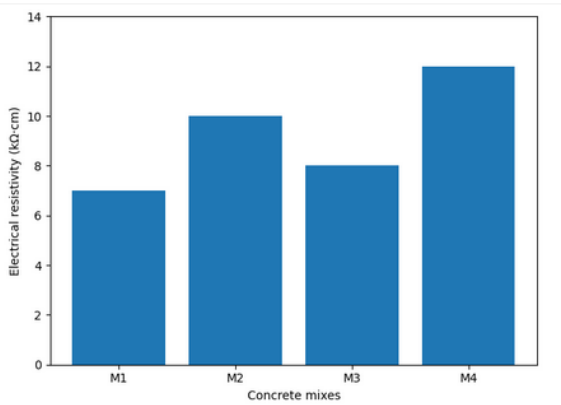


Fig. 6. Electrcial resistivity of cracked concrete (KO-cm).

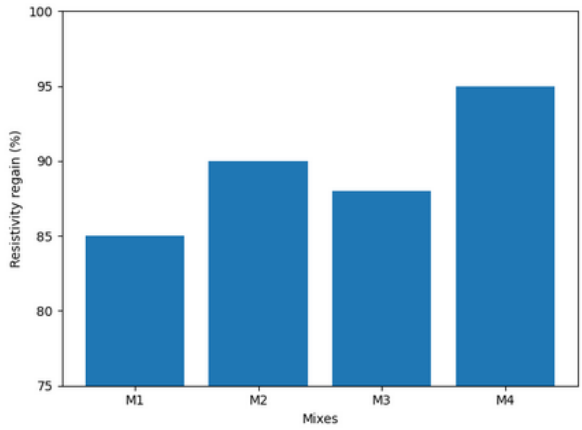


Fig. 7. Resistivity recovery ratio R(r) (%)

Electrical resistivity was calculated using Eq. (3):

$$\rho = \frac{A}{L} \times Z \qquad (3)$$

where $\rho$is the electrical resistivity (Ω·cm), $A$is the cross-sectional area (cm²), $L$is the specimen length (cm), and $Z$is the measured impedance (Ω).

Electrical resistivity of cracked specimens for mixes M1, M2, M3, and M4 was measured at 28 days. Compared with the cracked control mix (M1), the specimen containing 2% crystalline admixture (M4) showed a 72.90% increase in resistivity. Corresponding increases of 35.47% and 14.80% were observed for the PVA fiber mix (M2) and ECBP mix (M3), respectively, as shown in Fig. 6.

The resistivity recovery ratios of cracked specimens relative to the corresponding uncracked specimens were 84.53% (M1), 90.49% (M2), 89.25% (M3), and 94.43% (M4), as presented in Fig. 7. The increase in resistivity is associated with matrix densification due to continued hydration and the formation of pore-blocking products, particularly in specimens containing crystalline admixture . In general, electrical resistivity increased with curing age as hydration progressed and the cement matrix hardened.

3.3. Morphology of Healed Cracks

White sealing products formed within cracks after 42 days of water immersion, particularly in CA specimens (Fig. 8), consistent with previous reports . Specimens M2 and M4 exhibited greater crack healing than M1 and M3, where sealing was limited. These results confirm the effectiveness of CA in enhancing crack sealing within 42 days, in agreement with earlier findings . Overall, CA specimens showed superior crack-closing performance, especially under water immersion.

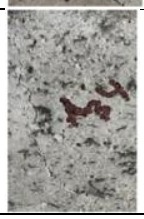
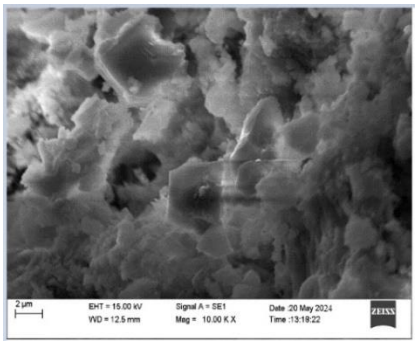
Mixes	**Early age crack**	
	3 days (before healing)	**42 days (after healing)**
M1		
M2		
M3		
M4		

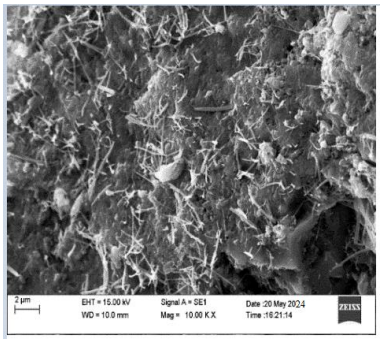
Fig 8. Self Healing of early age cracks

3.4.1. Scanning electron microscopy (SEM)

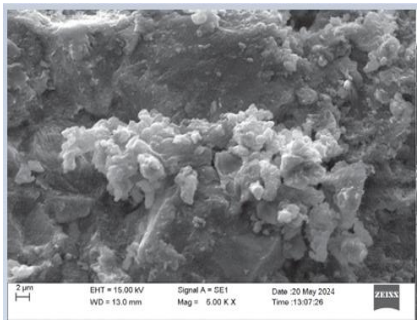
SEM examination of crack surface fragments from mixes M1–M4 after 42 days of water curing revealed hydration products such as C–S–H, calcite, and ettringite (Fig. 9). The M4 mix exhibited a denser and more compact microstructure than M1–M3, attributed to calcium carbonate formation induced by the crystalline admixture. Similar self-healing products have been reported in previous studies .



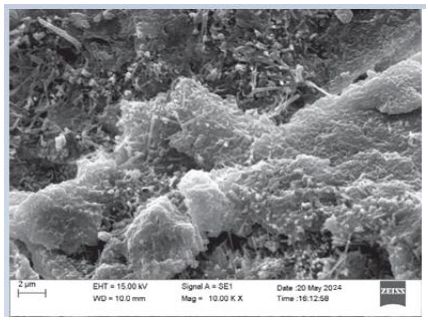
(a)M1



(b)M2



(c) M3



(d)M4

4. Conclusion

This study examined the self-healing behavior of early-age cracked concrete containing crystalline admixture (CA), ECBP, and PVA fiber. The results indicate that CA provides superior crack closure and recovery of mechanical and durability properties compared to ECBP and PVA fiber. Complete surface crack sealing was achieved in the CA mix after 42 days of water curing. Microstructural observations confirmed the formation of calcium silicate hydrate and calcium carbonate, with pronounced calcium carbonate precipitation in CA specimens. Overall, the incorporation of crystalline admixture significantly enhances the self-healing efficiency of concrete.

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